Yaojia Wang

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Degree: B.Eng.

Focus: HPC / Heterogeneous Computing / AI Acceleration / EDA + Architecture



Education

Fudan University	*B.Eng. in Microelectronics Science and Engineering*	2022.09 - 2026.06
Fudan University	*Ph.D. Student in Integrated Circuit Science and Engineering*	2026.09 - Present
University of California, Santa Barbara	*Exchange Student, Electrical Engineering*	2024.09 - 2024.12

- **GPA:** Undergraduate 3.58/4.0; exchange program 3.92/4.0
- **Selected Coursework:** Data Structures and Algorithms (A), Computer Architecture (A), Principles of Digital Integrated Circuit Design (A-), Fundamentals of Digital IC Design Automation (A), EDA System Software Analysis and Design Methodology (A-)
- **Honors and Awards:** Outstanding Graduate of Fudan University, Outstanding Student of Fudan University 2023, Outstanding Communist Youth League Member of Fudan University 2023, Second Prize in Contemporary Undergraduate Mathematical Contest in Modeling, Rongchang-Fudan Innovation Scholarship, Fudan University Second-Class Scholarship

Research Experience

Frontier-Based Joint Covering for Dual-Output LUT Mapping 2025.09 - 2026.04

- **Paper and submission:** First author; submitted the work to **International Workshop on Logic & Synthesis (IWLS) 2026**, selected as a **Best Student Paper Candidate**, and presented the work on site; **ISFPGA 2027** under review.
- **Algorithm and implementation:** Proposed **Frontier-Based Joint Covering (FJC)**, which moves cut covering and pairing for dual-output LUTs into frontier-level joint decision-making; implemented the `lut6d_map` pass, `GTP_LUT6D` technology-library cell, and automated evaluation scripts based on **Yosys/ABC**.
- **Experimental validation:** Completed equivalence verification on 20 EPFL benchmarks; reduced the total LUT count from 112,050 in the ABC baseline to 70,964, a **36.67%** reduction; further reduced LUT count by **2.97%** over the best-cut single-output + post-packing baseline.

Stabilizing and Optimizing Inter-Shell Routing in LEO Networks with Integrated Routing Cost 2024.12 - 2025.04

- **Paper and publication:** First author; led model design and manuscript writing; published the paper at IEEE/CIC International Conference on Communications in China 2025 (ICCC Workshops) and presented the work on site.
- **Algorithm and implementation:** Built a multi-orbit-layer (MEO/LEO) satellite-network simulation scenario and computed inter-layer link lengths and link switching frequencies.
- **Experimental validation:** Designed and ran routing-strategy optimization experiments comparing the traditional shortest-path algorithm, Adaptive Path Routing Scheme, and a new integrated routing-cost method with length and switching-frequency weights; validated transmission-performance improvements, reducing switching rates by 39.1% and 22.0%, respectively.

Projects

LLVM IR-based High-level Synthesis 2024.03 - 2024.05

- **End-to-end workflow:** Designed the repository structure and automation flow; implemented a pipeline from LLVM IR parsing, CDFG construction, resource-constrained scheduling, and Verilog generation to waveform-based validation; managed examples, tests, scripts, and project documentation (*GitHub*).
- **Algorithms and resource optimization:** Enhanced CDFG generation, optimized scheduling under resource constraints, designed register merging for better utilization, and built an automated testbench generator supporting scalar and array inputs.

Verilog-based RISC-V Multicycle CPU Design and Verification 2024.11

- **Architecture implementation:** Designed and implemented a 32-bit RISC-V multicycle processor supporting common arithmetic, logic, memory, branch, jump, and LUI instructions; integrated controller, datapath, ALU, register file, immediate extension, and unified instruction/data memory interface.
- **Control, datapath, and verification:** Implemented the main FSM across Fetch, Decode, memory, execute, write-back, branch, jump, and LUI states; verified PC, instruction, FSM state, ALUControl, source operands, and ALUResult with testbench waveforms; debugged the LUI write-back path and state transition to satisfy timing constraints.

MNIST-CLIP Multimodal Contrastive Learning Model 2025.05 - 2025.06

- Implemented a simplified CLIP-style model in PyTorch with a ResNet-based image encoder and a text-label encoder, mapping handwritten digit images and digit labels into a shared embedding space.
- Implemented bidirectional image-to-text/text-to-image contrastive loss, training-loss visualization, image classification, and similar-image retrieval; achieved **98.28%** classification accuracy on the evaluated MNIST set.

Fudan Institute for Spatial Internet Undergraduate Intern 2024.09 - 2025.06

- **Plotinus satellite-network digital twin system:** Built a Node.js-based satellite topology visualization interface for dynamic LEO satellite demonstrations; integrated backend data APIs and dashboards for link latency and load metrics; developed a basic topology configuration tool for satellite-node parameters and routing-path setup; received midterm excellence recognition in the Tengfei Innovation Program.

Technical Skills

- **Programming and Engineering:** Experienced with **C/C++**, **Python**, **Verilog**, and Shell; skilled in Linux debugging, scripting automation, data processing, and Git/Makefile-based workflows; familiar with JavaScript/Node.js.
- **Architecture and Performance:** Solid foundation in computer architecture, CPU datapath/control logic, RTL implementation, and simulation-based verification; interested in HPC, heterogeneous computing, and **AI hardware acceleration**; currently strengthening CUDA programming, GPU execution models, and performance analysis methods.
- **EDA and Digital Design:** Familiar with RTL design, logic synthesis, technology mapping, FPGA LUT mapping, and equivalence verification; used **Yosys/ABC**, Vivado, and related tools for algorithm implementation, resource optimization, and automated validation.
- **Algorithms, Models, and Experiments:** Experienced in graph algorithms, search, and resource-constrained optimization; familiar with **Transformer/LLM** fundamentals, prompt engineering, RAG/vector retrieval, and small-scale PyTorch/API experiments; able to build benchmark automation, result analysis, and ablation workflows.